

Zihao Zhang

Master's Student at Fudan University · Advisors: Prof. Yu-Gang Jiang & Prof. Zuxuan Wu
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RESEARCH PROFILE

Researcher at the intersection of **embodied intelligence** and generative visual modeling. I develop controllable and efficient video models for camera-aware scene evolution, large-motion temporal prediction, and human-motion editing—visual foundations for embodied agents. First/co-first author at CVPR 2025 and ACM MM 2026, with seven publications across CVPR, ICCV, ACM MM, SIGGRAPH Asia, and ICASSP.

RESEARCH THEMES

Generative World Models Camera- and motion-conditioned video synthesis for modeling how scenes evolve under embodied viewpoints and interactions.	Spatial & Motion Intelligence Multimodal spatial reasoning, camera planning, large motion, and temporally coherent scene understanding.	Efficient Visual Generation One-step pixel diffusion and lightweight control for scalable, high-resolution visual-dynamics simulation.
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EDUCATION

Sep 2023 — Jun 2026	Fudan University (QS WUR 2027: #26) Master's Degree · Shanghai, China School of Computer Science · GPA: 3.5 · Advisors: Prof. Yu-Gang Jiang & Prof. Zuxuan Wu
Sep 2018 — Jun 2022	Wuhan University (QS WUR 2027: #165) Bachelor's Degree · Wuhan, China School of Chemistry and Molecular Sciences

AWARDS & HONORS

2025	First-Class Academic Scholarship · Fudan University
2020 & 2021	First-Class Scholarship and Grant · Wuhan University
2025	EDEN Patent Application · Huawei Noah's Ark Lab · Large-motion video frame interpolation

RESEARCH EXPERIENCE

Apr 2025 — Mar 2026	Research Intern · Bilibili Inc. Virtual Human Group · Image and Video Generation · Shanghai, China SPEED: One-Step Pixel Diffusion for High-quality Video Frame Interpolation — Co-first author, ACM MM 2026 - Led the one-step pixel-space diffusion framework and designed a progressive multi-stage DiT, Noise-Update-Only Attention, and Drift-aware Timestep Sampling to learn motion, structure, and appearance from coarse to fine. - Achieved state-of-the-art visual-dynamics prediction: 8.8% lower LPIPS on SNU-FILM, 63.3% faster inference, 10.6% less memory, and up to 51.5% lower LPIPS on 4K benchmarks. CT-1: Vision-Language-Camera Models for Camera-Controllable Video Generation — Co-first author, ACM MM 2026 - Co-developed a Vision-Language-Camera model that grounds visual observations and language instructions in SE(3) camera trajectories, enabling spatially aware and temporally stable viewpoint planning. - Contributed to the CT-200K curation pipeline (>47M frames) and controllable video-diffusion system, improving camera-control success by 25.7% and advancing spatially grounded generative world modeling.
Jul 2024 — Mar 2025	Research Intern · Huawei Noah's Ark Lab · Image and Video Generation · Advised by Hang Xu · Shanghai, China EDEN: Enhanced Diffusion for High-quality Large-motion Video Frame Interpolation — First author, CVPR 2025 - Led EDEN's architecture and training, developing a Transformer Tokenizer, dual-stream temporal attention, and frame-difference conditioning for large, nonlinear motion. - Reduced LPIPS by nearly 10% on DAVIS and SNU-FILM and by about 8% on DAIN-HD, improving high-fidelity temporal state prediction under fast scene changes. MotionFollower: Editing Video Motion via Lightweight Score-Guided Diffusion — Third author, ICCV 2025 - Contributed lightweight pose/reference controllers and two-branch score guidance to transfer target body motion while retaining subject identity, background details, and camera dynamics. - Reduced GPU memory by approximately 80% versus MotionEditor while improving videos with large camera motion and complex backgrounds, enabling efficient motion-conditioned human-scene synthesis.

Publications

- 01 **EDEN: Enhanced Diffusion for High-quality Large-motion Video Frame Interpolation**
Zihao Zhang, Haoran Chen, Haoyu Zhao, Guansong Lu, Yanwei Fu, Hang Xu, Zuxuan Wu
CVPR 2025
- 02 **SPEED: One-Step Pixel Diffusion for High-quality Video Frame Interpolation**
Zihao Zhang*, Haoyu Zhao*, Siqian Yang, Yidi Wu, Yudong Jiang, Zuxuan Wu
ACM MM 2026
- 03 **CT-1: Vision-Language-Camera Models Transfer Spatial Reasoning Knowledge to Camera-Controllable Video Generation**
Haoyu Zhao*, Zihao Zhang*, Jiaxi Gu, Haoran Chen, Qingping Zheng, Pin Tang, Yeying Jin, Yuang Zhang, Junqi Cheng, Zenghui Lu, Peng Shu, Zuxuan Wu, Yu-Gang Jiang
ACM MM 2026
- 04 **MotionFollower: Editing Video Motion via Lightweight Score-Guided Diffusion**
Shuyuan Tu, Qi Dai, Zihao Zhang, Sicheng Xie, Zhi-Qi Cheng, Chong Luo, Xintong Han, Zuxuan Wu, Yu-Gang Jiang
ICCV 2025
- 05 **VIDiff: Translating Videos via Multi-Modal Instructions with Diffusion Models**
Zhen Xing, Qi Dai, Zihao Zhang, Hui Zhang, Han Hu, Zuxuan Wu, Yu-Gang Jiang
arXiv:2311.18837, 2023
- 06 **AniME: Adaptive Multi-Agent Planning for Long Animation Generation**
Lisai Zhang, Baohan Xu, Siqian Yang, Mingyu Yin, Jing Liu, Chao Xu, Siqi Wang, Yidi Wu, Yuxin Hong, Zihao Zhang, Yanzhang Liang, Yudong Jiang
SIGGRAPH Asia 2025 Posters
- 07 **EVA: Enhancing Anime Video Generation via Reinforcement Learning**
Yidi Wu, Bingwen Zhu, Zihao Zhang, Siqian Yang, Lisai Zhang, Baohan Xu, Mingyu Yin, Yanzhang Liang, Yudong Jiang
ICASSP 2026